

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Sc. (Physics) Examination March-April 2018

PS714 Solid State Physics-I

Date: 29.03.2018, Thursday

Time: 10:00 a.m. To 10:20 a.m.

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

[10]

1. The regular period arrangement of atoms is known as

[1]

- (a) crystal, (b) quasi crystal,
(c) amorphous material, (d) glassy material

2. A hexagonal Bravais lattice has lattice parameters:

[1]

- (a) $a = b = c, \alpha = \beta = \gamma = 90^\circ$ (b) $a = b, c, \alpha = \beta = \gamma = 90^\circ$
(c) $a = b, c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$ (d) $a \neq b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$

3. A glide operator consists of:

[1]

- (a) Reflection plus rotation (b) Rotation plus translation
(c) Reflection plus translation

4. For the Bragg reflection from a set of parallel adjacent planes separated by d the wavelength of the X-rays must be **[1]**
(a) Less than d (b) Greater than d
(c) Greater than $2d$ (d) Less than or equal to $2d$
5. Interatomic forces in solids are **[1]**
(a) attractive only (b) repulsive only
(c) neither attractive nor repulsive (d) both attractive and repulsive
6. In Debye's theory of specific heat of solids, the atomic oscillators obey **[1]**
(a) MB statistics (b) FD statistics
(c) BE statistics (d) All of the above
7. The spring constant α and Young modulus Y in solids are related by **[1]**
(a) $\alpha = Ya$ (b) $\alpha = Y/a$
(c) $\alpha = a/Y$ (d) $\alpha = Ya^2$
8. Lorenz number is a ratio of **[1]**
(a) Electrical conductivity to thermal conductivity
(b) Thermal conductivity to electrical conductivity
(c) Thermal resistivity to electrical resistivity
(d) None of the above
9. Fermi distribution function reduces to Boltzmann distribution function in the limit of **[1]**
(a) high temperature, (b) low temperature,
(c) high energy, (d) low energy
10. The knowledge of lattice vibration is useful for providing information about **[1]**
(a) specific heat (b) thermal expansion
(c) thermal conductivity (d) all of the above

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PS714 Solid State Physics-I

Date: 29.03.2018, Thursday

Time: 10:20 a.m. To 12:00 p.m.

Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q - II

10

1. Why FCC metals are ductile while BCC and HCP metals are not? 2
2. Using structure factor F_{hkl} , show that which of the following reflections would be missing in a bcc lattice: (100), (110), (111), (200) ? 2
3. A Cu target emits an X-ray line of wavelength $\lambda = 1.54 \text{ \AA}$. Knowing that the density and atomic weight of Al are, respectively, 2.7 g/cm^3 and 27 g/mol , compute the value of Avogadro's number. 2
4. Draw the dispersion relation for linear diatomic lattice showing acoustic and optical modes with appropriate boundary conditions. 2
5. The atomic radius of Na is $1.86 \text{ \AA}$. Calculate the Fermi energy of sodium at absolute zero. 2

SECTION – II

Q-III

15

1. Diamond has fcc structure in which the basis is composed of two identical atoms, one at the lattice point and another at a point $(a/4, a/4, a/4)$ relative to the first atom, where a is the edge of the cube. Find the geometrical structure factor for diamond and then show which of the reflections from (100), (110), (111), (200), (210), (220), (211) is missing in diamond? **4**
2. Show that the reciprocal lattice of an fcc Bravais lattice with conventional cubic cell side a is a bcc lattice with conventional cubic cell side $4\pi/a$. **4**
3. The longitudinal and transverse velocities of sound in diamond, a cubic crystal, along the [100] direction are, respectively, 1.76 and 1.28×10^6 cm/s. The longitudinal velocity in [111] direction is 1.86×10^6 cm/s. From these data and the fact that the density is 3.52 g/cm^3 , calculate the elastic constant C_{11} , C_{12} , and C_{44} for diamond. **3**
4. Silver (fcc) has an atomic radius of $1.44 \text{ \AA}$. Assuming silver to be monovalent metal, calculate the value of the Fermi energy, the Fermi temperature and the Fermi velocity. **4**